Shopee Price Match Guarantee Algorithm based on multimodal learning

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Abstract— Shopee has been a popular online shopping website in the Southeast Asia. Customers appreciate its easy, secure, and fast online shopping experience tailored to their region. At the same time, it allows customers to choose the one with the lower price of the same product. It relies on the product matching, that is the same product with the same description image must be removed. The base technology to achieve this function is multimodal learning, in which we focus on the images and text. In our article, we proposed a new multimodal learning model mainly based on transformer and BERT. For image matching, we use NFNet, Swin_Transformer and Efficientnet to get image embeddings. For text matching, we use Distil-Bert, Albert, Multilingual Bert and TF-IDF to get text embeddings. After we get the embedding vector, we choose KNN to classify. We use cosine and distance to measure the similarity of the different models. It is worth mentioning that the loss function is Arcface, not the traditional Softmax, which improve the difficulty of training to ensure the final performance in the test periods. In addition, 7 models vote for the final results ensuring the effect of prediction. To avoid the bad matching result，we add some post-processing process.

Index terms: Shopee, multimodal learning, voting, post-processing, Arcface

I. INTRODUCTION

With the development of technology, in many countries, people tend to shop online, such as Amazon in America and Taobao in China. There are hundreds of millions of consumers active in the online shopping software. Shopee is a popular software in south-east Asia and its and one of its important characteristics is that it allows many retailers to sale the same products. [1] It also provides strong payment and logistical support along with a 'Lowest Price Guaranteed' feature on thousands of Shopee's listed products. Due to this, savvy shoppers could have more choices for buying products in a lower price in the shopping periods. On the other hands, retail companies use a variety of methods to ensure their products are the cheapest to attract customers to buy. Two different images of similar goods may represent the same product or two completely different items. Retailers want to avoid misrepresentation and other problems arising from confusing two different products and avoid affecting the buyer experience by multiple displays of the same item. What Shopee needs to do is do products matching to find and clear duplicate product information.

Now Amazon has used the function of the products matching for improving the user experience. Merchants can match the item they want to sell with an existing item by matching the item. It adds your quote information to the item entry. If the item does not already exist in the catalog, the merchant can create a new product details page for it. This matching function is based on the multimodal learning. [2] We need to build image models, text models and combine everything to judge whether the product matching. Thanks to Kaggle for providing us with the Shopee's dataset. The whole of our model is using Pytorch framework.

The structure of this paper is as follows: in the section II, we introduce some related work about multi-model learning; and next section III, the methodology of our article is shown. In the section IV, we introduce the experiments process and it also contains the datasets, feature engineering and metrics. Finally, section V, we conclude our work and propose the plan in the future.

II. Related Work

The products matching problem could be converted into the image text matching problem.

A rich line of studies has explored mapping whole images and full sentences to a common semantic vector space for image-text matching [3]. Kiros et al. [4] made the first attempt to learn cross-view representations with a hinge-based triplet ranking loss using deep Convolutional Neural Networks (CNN) to encode images and Recurrent Neural Networks (RNN) to encodes sentences. Faghri et al. [5] leveraged hard negatives in the triplet loss function and yielded significant improvement. Peng et al. [6] and Gu et al. [7] suggested incorporating generative objectives into the crossview feature embedding learning.

- *Our Contribution*

✓ A new multimodal proposed by us. It contains seven sub-models about image and text matching.

✓ F-1 score is the metrics in our experiments to evaluate our model's performance.

✓ We utilize the post-processing and voting to solve the problem that there are little number of matching samples.

✓ The loss function we use is the Arcface to increase the difficulty of training which could guarantee the performance in the test periods.

III. Methodology

The commodity matching problem can be converted into a two-category problem, and the output probability is similarity.

The whole structure of our model is shown in the following figure 1.

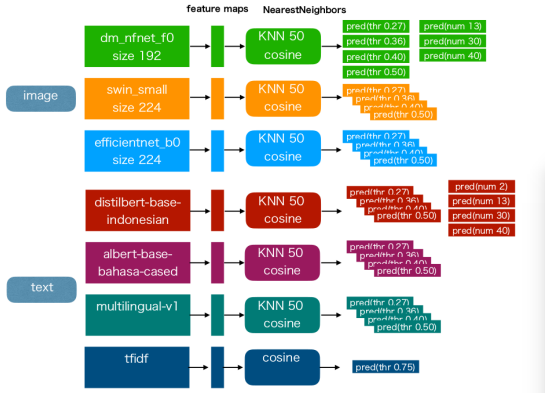


figure 1: the structure of our model

For image matching, we use NFNet, Swin_Transformer and Efficientnet to get image embeddings. For text matching, we use Distil-Bert, Albert, Multilingual Bert and TF-IDF to get text embeddings. We will briefly introduce these models.

- Image matching

➢ NFNet [8]

NFNet is not a normalized ResNet network. The NFNet's contribution is as follows:
1) An adaptive gradient trimming (Adaptive Gradient Clipping, AGC) method is proposed to cut the gradient based on the unit ratio of the gradient model to the parameter model. AGC can train larger, large-scale data-enhanced, non-normalized networks.
2) A new network called Normalizer-Free ResNets was designed, which achieves the highest level of training wait times over a wide range of imageNet validation sets.
3) The NFNet-F1 model achieves accuracy similar to that of the EfficientNet-B7, while the training speed is increased by 8.7 times, while the largest version of the NFNet model sets a new LEVEL of SOTA, achieving top-1 accuracy of 86.5% without additional data.

➢ Swin Transformer [9]

Swin mimics CNN's feature layer design from large to small, giving swin a feature similar to CNN's Sense of the Wild.
The entire model, as described in the figure above, consists of four Stages, each of which reduces the resolution of the input feature map and expands the feel of the wild layer by layer, like CNN.

After entering the original picture, Swin performs a Patch Embedding operation, cutting the picture into tiles and embedding it in Embedding, a step similar to that in vit.
Each Stage contains Patch Merging and multiple Blocks. The Patch Merging module mainly reduces the picture resolution at the beginning of each Stage to achieve a wild-feeling structure. Block's specific structure, shown on the right, consists mainly of Layer Norm, MLP, Window Attention and Shifted Window Attention.

➢ EfficientNet [10]

To improve the accuracy and speed of your network, you need to balance the three dimensions of width, depth, and resolution. So how to balance these three dimensions at the same time can get the best network architecture under the limit conditions, the previous method is basically one or two of these three dimensions to parameterize, through experiments to determine the best network structure, obviously the best hand-obtained is likely to be local optimal, so EfficientNet through neural network architecture search (NAS) way to balance the three dimensions at the same time, search to get the optimal network architecture.

- Text matching

➢ Distil-Bert [11]

Knowledge distillation is a common method of model compression, unlike pruning and quantification in model compression, knowledge distillation is to train a small model by building a lightweight small model, using the supervision information of a large model with better performance, in order to achieve better performance and accuracy. The overall structure of the Studio model in DitilBert is the same as Bert's, but the Studio model removes token-type embedding and pooler while the number of model layers is halved. The author optimizes the matrix calculation of the layer normalization and the full connection layer in transformer. At the same time, the author notes that the change of hidden size dimension has less effect on the efficiency of model calculation than the change of layer number, so it mainly pays attention to the influence of model layer number on calculation efficiency. The authors use layers in the Teacher model to initialize the Studio model, and the training method is consistent with Roberta's.

- Albert [12]

Albert mainly made 3 improvements to BERT, reduced the overall number of parameters, speeded up the training speed, increased the model effect. Three improvements are: 1) Factored embedding parameterization, 2) Cross-layer sharing parameter, 3) Inter-sentence coherence loss.

- Multilingual Bert [13]

Because Shopee is widely used in South-East Asia, it has many kinds of language. The core idea of Multilingual BERT is to handle all target languages with the same models and weights. First, because different languages have different vocabulary, Multilingual BERT has built a glossary that contains all the target languages. Second, the Multilingual BERT encoding section is the same structure as the BERT and is shared across all languages. The pretrained task is still BERT's masked language model. Each of these samples is a monolingual paragraph, with all languages trained alternately in a multitasking manner.

- TF-IDF [14]

TF-IDF is the shortage of Term Frequency-Inverse Document Frequency to assess how important a word is to a file set or one of the files in a corpus. TF means Term Frequency, it is usually normalized to avoid the tendency to longer files. IDF means inverse Document Frequency, indicating the prevalence of keywords. If there are fewer documents containing the entry i and the larger the IDF, the term has good category differentiation. The IDF of a particular word can be divided by the total number of files by the number of files containing the word, and then the resulting commercial icing on the basis of the resulting

IV. EXPERIMENTS

In this section, we introduce our Shopee datasets, and the feature engineering. In addition, we describe our metrics (F-1 score) and some training tricks.

A. *The dataset*

Kaggle platform provides 32.4k number of posts as the training dataset. Each row contains the data for a single posting. Multiple postings might have the exact same image ID, but with different titles or vice versa. We have five columns: posting_id, image, image_phase, title, label_group. Posting_id means the ID code for posting. The meaning of different columns is shown in following table 1.

posting_id	the ID code for the posting
Image	the image-id/md5sum.
image_phash	a perceptual hash of the image
title	the product description for the posting.
label_group	ID code for all postings that map to the same product.

table 1: the meaning of different columns

B. *Metrics*

In our task, we use F1 Score as our metrics to evaluate our model's performance. The definition of F1 Score is as follows:

$$F1\ Score = \frac{2 \times (Precision \times Recall)}{Precision\ +\ Recall}$$

It is a combined metrics based on precision and recall. And the definition of the precision and recall is as follows:[10]

$$Precision = \frac{TP}{(TP + FP)}$$

$$Recall = \frac{TP}{(TP + FN)}$$

C. *Training Tricks*

1) Loss function--- Arcface

In order to enhance the classification ability of softmax, in fact, it is necessary to achieve two points in the distribution: make the distance between the same class closer and make the distance between different classes longer. Therefore, in our article, we choose Arcface as the loss function to improve the difficulty of training and then get the better performance in the test process.

2) Voting method

Because our whole model contains 7 sub-models, they give 7 prediction result. If there are 5 or 5above models all determine the same one category of the product, we classify the product into the category.

3) Post-processing

According to the prediction results of the optimal model in the early stage, our statistical results found that there are a large number of unmatched samples, that is, the matching result is only its own, so the matching standard can be lowered (increasing the matching distance threshold or reducing the similarity threshold). At this time, we continue to adjust the threshold until the requirements are met. The following figure 2 show the training process.

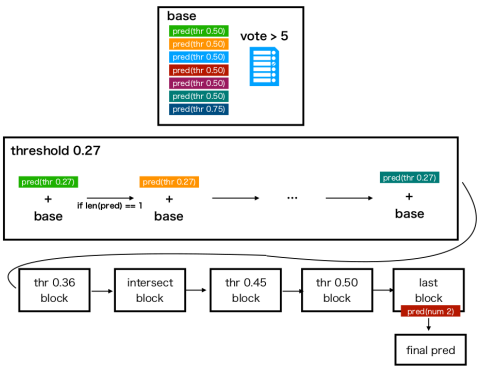


figure 2: training process

V. CONCLUSIONS

In this part, we conclude our work. Shopee has been a popular online shopping website in the Southeast Asia. Customers appreciate its easy, secure, and fast online shopping experience tailored to their region. At the same time, it allows customers to choose the one with the lower price of the same product. In the section II, we introduce some related work about multi-model learning, such as RNN and CNN. And in the next section III, the methodology of our article is shown. We introduce the structure of the whole model, and it based on 7 sub-models. For image matching, we use NFNet, Swin_Transformer and Efficientnet to get image embeddings. For text matching, we use Distil-Bert, Albert, Multilingual Bert and TF-IDF to get text embeddings. We will briefly introduce these models. In the section IV, we introduce our datasets, the metrics and some training tricks. In the future, we will optimize our model form different aspects.

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